Energy Strategy Reviews 56 (2024) 101592

Contents lists available at ScienceDirect

Energy Strategy Reviews

journal homepage: www.elsevier.com/locate/esr

Environmental regulation, energy poverty and social welfare: Based on China's provincial panel data

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ARTICLE INFO

Handling editor: Mark Howells

Keywords:
Environmental regulation
Energy poverty
Social welfare
Principal components analysis
Extended linear expenditure system

ABSTRACT

Alleviating energy poverty, improving the ecological environment, and promoting social welfare have become important issues in society. By taking panel data from 30 provinces in China from 2003 to 2020 as the research object, fixed effects models and system-GMM models are adopted, and the impacts of environmental regulations on social welfare and the moderating effect of energy poverty are studied in this paper. The results show that enhanced environmental regulation has a significant negative impact on social welfare, while flexible environmental regulation has the opposite, especially in energy poverty regions. Moreover, energy poverty has a significant negative impact on social welfare, and energy poverty reduces the consumption capacity of clean energy, which may pollute the environment and damage health, thus reducing social welfare. Additionally, the energy poverty has significant positive moderating effect, that is, energy poverty aggravates the inhibitory effect of enhanced environmental regulation on social welfare, and promotes the impact of flexible environmental regulation on social welfare. Finally, energy poverty varies significantly in different regions, and the energy poverty lines in the eastern, central and western regions, as well as the national are 503.381, 442.758, 359.461 and 444.546, respectively.

1. Introduction

Sustainable development should not only provide adequate energy services, such as renewable energy including wind power and solar photovoltaic energy [1] has become one of the most important measures for China to achieve carbon neutrality goals [2], but also mitigate the adverse impact on the environment [3]. Environmental regulation plays an important role in reducing energy consumption, improving ecological environment and enhancing social welfare [4,5]. At present, the establishment of diversified energy supply systems such as coal, oil, natural gas, wind power and hydropower has alleviated China's energy shortage to a certain extent, but the contradiction of uneven energy distribution is still prominent, which is mainly reflected in energy poverty [6,7]. The scarcity of energy and the negative externality of environmental pollution have given birth to environmental regulation [8–10]. Although short-term government subsidies could alleviate the pressure of the energy crisis, in the long run, policies must be consistent with the goals of mitigating climate change [11] and sustainable development. Consequently, how to combine China's national conditions and the current situation of energy poverty, fully utilize different types of environmental regulations, and through incentives and constraints on enterprises, reduce pollution and improve environmental quality, is an important issue to achieve the improvement of social welfare levels in Life, Health, and Education.

To be specific, environmental regulation can effectively reduce the consumption of fossil energy [12] and carbon emissions, and thus improve social welfare [13]. Moreover, the environmental regulation can also have an impact on energy poverty. For example, the government supports the transformation of traditional coal-fired boilers to reduce the environmental pollution, popularize the use of renewable energy, and thus reduce energy poverty [14]. In addition, energy poverty can also affect social welfare. With the deepening of energy

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https://doi.org/10.1016/j.esr.2024.101592
Received 30 August 2023; Received in revised form 7 July 2024; Accepted 10 November 2024
Available online 20 November 2024
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poverty, the quality of life has also declined. Research shows that energy poverty will reduce the income capacity of a family by 48 % [15]. As a result of energy scarcity, many families do not have access to electricity and have to choose firewood and lighting oil as energy sources, which increases the risk of suffering from diseases. Therefore, it is urgent to reduce or even eliminate energy poverty.

Most existing studies used the system-GMM and two-way fixed effect models to examine the impact of environmental regulation on social welfare [16,17], while the quantile and logistic regression models are used to investigate the relationship between energy poverty and social welfare [18,19]. However, the environmental regulation affects social welfare through energy poverty are still not discussed. Based on the stakeholder theory, the relationship between environmental regulation, energy poverty and social welfare is analyzed in this paper. At the same time, the fixed effect and system-GMM models are used in this paper to explore the impact of environmental regulation on social welfare, and further examine the moderating effect of energy poverty. This paper constructs a new model framework of environmental regulation, energy poverty and social welfare at the theoretical level to eliminate energy poverty, achieve ecological progress, and improve social welfare.

The marginal contributions of this paper are in three aspects: Firstly, the environmental regulation, energy poverty and social welfare are all included in the same research framework, further enriching the existing literature on the relationship between environmental regulation and social welfare. Secondly, considering the heterogeneity of various environmental regulations, this paper respectively examines the impacts of enhanced environmental regulation and flexible environmental regulation on social welfare, and examines the moderating effect of energy poverty in two types of environmental regulations. Thirdly, this paper adopts the principal component analysis (PCA) method to comprehensively measure the level of social welfare. Meanwhile, the extended linear extension system (ELES) is used to calculate the energy poverty lines of the eastern, central, and western regions of China and the country, which helps to adopt reasonable environmental regulation policies for energy poverty in different regions and levels, reduce or eliminate energy poverty, and improve social welfare levels. In addition, this article may also have some shortcomings, the research content may be limited to the field of economics, and there may be a lack of comprehensive interdisciplinary research such as sociology and environmental science, which will be conducive to providing a more comprehensive perspective in the future.

2. Literature reviews

2.1. Environmental regulation and social welfare

There is a dynamic relationship between environmental regulation and social welfare. On the one hand, environmental regulation improves social welfare. The early use of EUCARS model to simulate the policy decision of reducing passenger transport emissions in Europe shows that appropriate environmental regulation could not only improve the environment, but also ensure the social welfare effects [20]. Moreover, carbon taxes have reduced environmental damage by nearly 50 % in all regions, including low-income countries, thus improving social welfare [21]. In addition, from the perspective of product cost, environmental regulation increases the production cost of conventional products in order to reduce the damage to the environment. For this reason, more strict environmental regulation on traditional products promotes the possibility of improving social welfare [22]. On the other hand, environmental regulations lead to a decline in social welfare. For consumers, the increase of carbon tax will reduce social welfare [23]. Moreover, environmental regulations reduce the wages of employees in polluting enterprises, for whom social welfare decreases [24].

Furthermore, making appropriate environmental regulations according to the actual situation could enhance the social welfare effect. Under the implementation of environmental regulations, the government should utilize market forces and take into account consumer preferences to improve social welfare [25]. In emerging countries, green tax incentives related to environmental sustainability, such as taxable income adjustments and ITC (Investment Tax Credits), are becoming increasingly popular. Green tax incentives to respond climate change are implemented, and stakeholders benefit from tax incentives to achieve the goal of sustainable social development [26]. If regions are divided, it could be found that environmental regulations in different regions would have different impacts on people's subjective well-being. Thus, the government should make appropriate environmental regulations according to local conditions to maximize people's welfare [27]. Similarly, the impact of subjective air pollution on happiness is also an aspect of social welfare. It is found that people's willingness to be happy is also very important in the process of improving air environment. Therefore, appropriate environmental regulations should be formulated to improve social welfare [28–30].

2.2. Environmental regulation and energy poverty

Environment and energy are also hot issues of world concern. Developing rational energy and environmental policies [31] in various relevant management and control processes to conserve energy [32] and mitigate the environmental impact of energy poverty remains a major challenge [33]. On the one hand, energy poverty will contribute to environmental degradation [34], which requires environmental regulation. For example, a study of the BRICS countries during 1989–2016 found that energy poverty exacerbated environmental stress in these countries [35]. At this time, environmental regulations are necessary, but strict environmental regulations will lead to a greater burden on households using dirty energy, and increase energy poverty [36]. On the other hand, climate change will further exacerbate energy poverty. A research of energy poverty in Vietnam found that heat shock can reduce agricultural output and income, and exacerbate energy poverty [37]. At the same time, some scholars study the consistency of trade openness in encouraging universal use of green energy, and the links between environmental degradation, economic development, oil prices and green energy consumption [38]. Environmental degradation has a negative impact on green energy consumption, while oil prices and economic development have a positive impact on green energy consumption.

Besides, there are many researches on the relationship between environment and energy from the aspect of building energy. High energy consumption in the building industry, energy poverty and environmental change are the three major problems facing the European built environment. The connection, synergy and influence among the three issues of energy consumption of buildings, energy poverty and local climate change are comprehensively analyzed [39]. Since building energy consumption decreases, energy poverty decreases, and the corresponding emissions decrease, the impact on the environment should be reduce. On the contrary, if the environmental temperature rises, the energy consumption for cooling will increase, and the degree of energy poverty will rise [40]. Consequently, energy renovation of existing buildings is an effective action to improve energy poverty and reduce greenhouse gas emissions [41,42]. In addition, studies on energy poverty and carbon intensity in the construction industry have found that energy poverty and carbon intensity have a positive impact, and technology level plays a moderating role in the effect of energy poverty and carbon intensity. Thereby, technological innovation is also one of the effective ways to improve energy poverty and environmental change [43]. For example, the energy storage technologies (ESTs) play a vital role in ensuring energy security and addressing the challenges posed by climate change, which allow us to overcome the intermittent and mismatch between energy supply and demand caused by unpredictable nature of renewable energy [44].

2.3. Energy poverty and social welfare

Inefficient energy use may exacerbate energy poverty and ultimately reduce social welfare. For example, energy poverty is widespread in India, where inefficient use of biomass fuels may cause health hazards and reduce social welfare [45]. Additionally, long-term energy poverty reduces the social welfare of a country, and only energy efficiency could benefit from it. According to the survey, power consumption ranks second among energy poverty, and the lack of access to modern electricity in regions with low energy efficiency may exacerbate energy poverty [46]. In addition, household energy poverty is closely related to income level, and unaffordable energy has a huge impact on people's health and other welfare. Promoting access to and use of clean energy in households and alleviating energy poverty have been at the heart of worldwide efforts [47,48]. A study on the incidence of energy poverty and people's social welfare in Cambodia found that energy poverty is closely related to income consumption power, and therefore household well-being is greatly affected. The study suggests that relevant policies should be developed to reduce energy poverty by providing clean and affordable energy to poor households [49–51]. According to the investigation of five small villages in Jinzhong City, Shanxi Province, China, energy poverty is manifested one by one, and low income is one of the reasons. Attention to and alleviation of energy poverty is conducive to improving public happiness and promoting ecological progress [52].

At the national level, energy poverty affects the economies of both developed and developing countries, and its mitigation has important implications for social welfare [53]. In the investigation of energy poverty in Ghana, it is emphasized again that energy poverty is one of the major obstacles to national development, and found that high welfare level will reduce the probability of household energy poverty [54,55]. Similarly, the conclusion has been drawn that energy poverty is harmful to people's health by analyzing and comparing urban and rural areas. Moreover, multidimensional energy poverty reduction could be applied to reduce regional disparities and improve social welfare [4,56,57].

2.4. Environmental regulation, energy poverty and social welfare

Energy poverty is a complex issue that affects not only the use of energy, but also the inefficient use of energy, which would have an impact on the environment, release more carbon dioxide and affect social welfare. As a result, appropriate policies should be formulated to alleviate these series of problems. For example, the model of household energy use and air pollutant emission in Nepal is analyzed, and it is found that the local indoor pollutant emission will continue to increase in the future. The energy development index is designed to evaluate energy access and poverty, and the existence of energy poverty is found. It is suggested that the government adopts more appropriate policies to promote alternative energy for households [58]. Similarly, the geographical and socio-economic situation of Rwanda affects the energy situation and seeks energy and environmental solutions from an environmental or energy perspective [59]. Inspecting the EU member states at the macro level, scientists are called on that not only to further study energy poverty, but also to participate in the formulation of effective policies to alleviate energy poverty and make the transition to green, healthy and sustainable [60].

Furthermore, poverty eradication and climate change are two major challenges facing ''The Times'' [61]. Taking Greece as a case study, it emphasizes that energy poverty is not only a big problem, but also a complex one, which will have a serious impact on the social environment, welfare level and even the economic level of the whole society [62]. Consequently, it is necessary to solve the problem of energy poverty. Studies have found that when energy prices rise [63], economies choose more energy-efficient technologies. This would not only contribute to rapid economic growth and improve social welfare, but also reduce energy consumption and carbon emissions [64], making a great contribution to alleviating energy poverty and climate problems [65].

Therefore, environmental regulation has both direct and indirect effects on social welfare. The stakeholder theory holds that businesses should be accountable not only to shareholders but also to all other stakeholders affected by corporate activities, such as employees, suppliers, local communities, and creditors [66]. In managing an organization, moral and ethical considerations should be taken into account, such as those related to corporate social responsibility, market economy, and social contract theory. Stakeholder theory posits that the success of a business lies in meeting the needs of all stakeholders, not just the interests of shareholders. Some scholars have studied the impact of corporate governance on social sustainability and the distance to default of Sukuk issuers to explore the mitigating effect of these variables and the mediating role of corporate social sustainability [67]. Based on the stakeholder theory, there is a literature on the impact of accounting number game on the financial sustainability of enterprises, and external block-holders and institutional ownership are used as moderating variables [68]. From the perspectives of governments, enterprises, and individuals, the impact of heterogeneous environmental regulations on social welfare are studied in this paper, taking energy poverty as a moderating variable.

3. Mechanism

The existence of energy poverty is not conducive to the improvement of social welfare, while environmental regulation can significantly improve social welfare. In order to show the relationship between them more intuitively, Fig. 1 has been drawn in this paper.

Stakeholder theory is a theory of organizational management and business ethics. It can be applied to analyze how environmental regulation affects social welfare by influencing different stakeholders, including the government, the public, and enterprises. For example, the government promotes the construction of new energy infrastructure and the promotion of clean energy by strengthening environmental regulations. This not only helps to ensure energy supply and provide new employment opportunities but also promotes local economic development, thereby enhancing social welfare. As active participants in environmental protection, the public can help improve social welfare by reporting the behavior of polluting enterprises to the government and the media, prompting enterprises to reduce pollution emissions, stimulate technological innovation, improve the local ecological environment, and enhance public happiness.

The environmental regulations are divided into enhanced environmental regulation and flexible environmental regulation in this paper. Firstly, the enhanced environmental regulation might influence social welfare through "crowding-out effect" and "innovative compensation effect". Specifically, in the early stage of implementing enhanced environmental regulation, the intensity of regulation is weak, and the environmental costs faced by producers and operators are greater than the benefits of innovation compensation, which hinders the green transformation of economy and is not conducive to the improvement of social welfare. But with the rationalization of regulation, the benefits of innovation compensation are greater than the environmental costs, which would help accelerate the green transformation and improve social welfare [25]. In addition, the government's enhanced environmental regulation has led the construction of new energy infrastructure and the promotion of clean energy, which has guaranteed energy supply, provided new employments and promoted local economic development [58,60]. At the same time, the promotion of clean energy has changed people's traditional energy use habits and lifestyles, increased the consumption demand for clean energy, and improved the energy utilization structure [65]. Therefore, the enhanced environmental regulation could significantly improve the level of energy poverty, and with the weakening of energy poverty and the improvement of local economic benefits, social welfare would also be greatly improved.

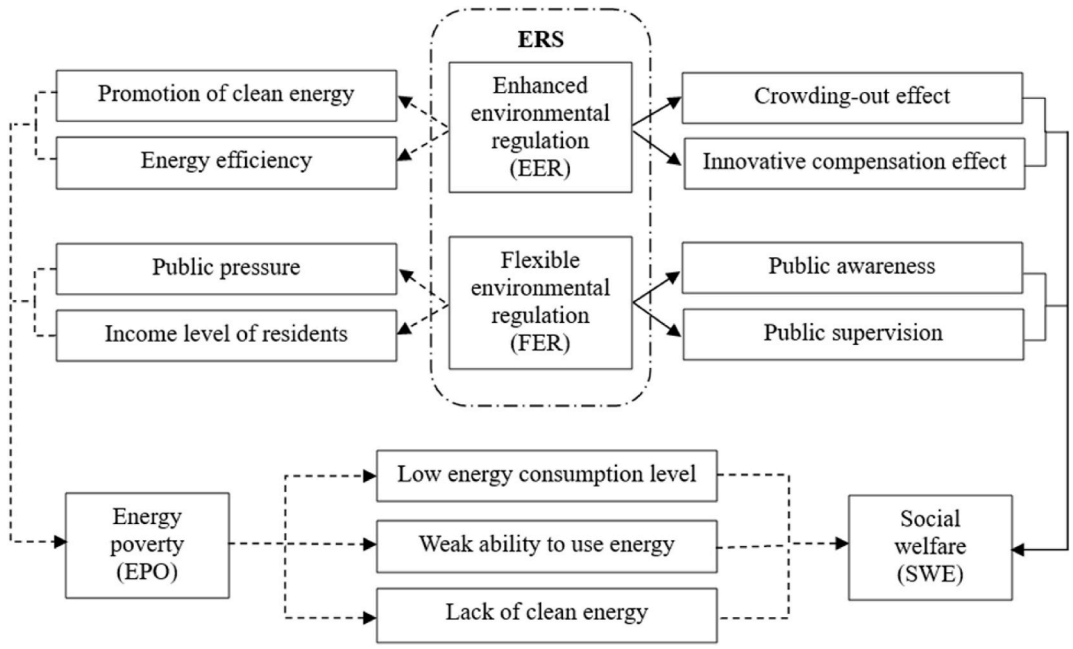

Fig. 1. The relationship between ERS, EPO and SWE.

Secondly, compared with enhanced environmental regulation, the flexible environmental regulation has less effect on social welfare. On the one hand, with the enhancement of public awareness of environmental protection and environmental supervision, for the pursuit of good quality of life, it would put pressure on polluting enterprises by reporting to the government and media exposure. At the same time, it would also spontaneously establish some non-profit environmental protection organizations to negotiate with polluting enterprises. Under the external pressure, enterprises would reduce pollution emissions, stimulate technological innovation, improve local ecological environment, enhance public happiness, and thus help to improve social welfare [29,30]. However, the public's demands maybe are not completely consistent with the government's needs. When the public puts too much pressure on enterprises, it would hinder the normal production and operation activities of enterprises, increase production costs and reduce profitability, which is not conducive to the improvement of social welfare. On the other hand, the flexible environmental regulation is a direct manifestation of the public's pursuit of high-quality life, and increases with income and education levels. With the increase intensity of flexible environmental regulation, the role of the public in environmental protection is more prominent, and the appropriate pressure of the public on polluting enterprises is conducive to technological innovation of enterprises, adjustment of energy structure, reduction of pollution emissions, promotion of green transformation of the whole society, and improvement of energy poverty and social welfare [43]. The study of the impact of different environmental regulations on social welfare under the intensity of energy poverty is of great significance for formulating effective energy policies and poverty alleviation interventions. It would help to address global energy poverty and improve the quality of life and welfare of residents.

4. Methodology

4.1. Model construction

In order to test the impact of the environmental regulations, energy poverty and social welfare, this paper uses the system generalized moment (sys-GMM) estimation method to build the panel data model. The sys-GMM allows random error items to have heteroscedasticity and serial correlation, so the parameter estimator obtained is more effective than other traditional parameter estimation methods, and can better deal with the autocorrelation and endogenous problems of variables. Some scholars have studied the impact of green economy on environmental sustainability based on the VAR lag sequence [68].

First of all, in order to study the impact of ERS on SWE, regression panel models are established, such as models (1) and (2):

$$SWE_{it} = \alpha_0 + \alpha_1 ERS_{it} + \alpha_2 CONTROL_{it} + \mu_{it} \tag{1}$$

$$CONTROL_{it} = IS_{it} + TI_{it} + FP_{it} + ED_{it} \tag{2}$$

SWE and ERS are explained variable and explanatory variable respectively. SWE stands for social welfare. ERS stands for environmental regulations. i represents the province and t represents the year. In order to control other influencing factors of social welfare, the control variables selected in this paper include industrial structure (IS), technical innovation (TI), financial expansion (FP), and education (ED). α_0 is a constant term, α_1 and α_2 are the coefficients of explanatory variables and control variables respectively. μ_{it} is a random disturbance term.

Secondly, the energy poverty (EPO) is introduced in this paper to study the moderating effect in the impact of ERS on SWE. Based on the influence mechanism of ERS and EPO on SWE (as shown in Fig. 2), the measurement model has been scientifically constructed. Model (3) explains the relationship between ERS and SWE. In model (4), $ERS_{it} \times EPO_{it}$ represents the moderating effect of energy poverty on environmental regulations affecting social welfare. β_0 and γ_0 are constant terms. β_1 and γ_1 are the coefficients of the independent variables, γ_2 represents the coefficient of the interaction term between ERS and EPO. β_2 and γ_3 are the coefficients of the control variables. φ_{it} and θ_{it} are random disturbance terms.

$$SWE_{it} = \beta_0 + \beta_1 ERS_{it} + \beta_2 CONTROL_{it} + \varphi_{it} \tag{3}$$

$$SWE_{it} = \gamma_0 + \gamma_1 ERS_{it} + \gamma_2 EPO_{it} + \gamma_3 ERS_{it} \times EPO_{it} + \gamma_4 CONTROL_{it} + \theta_{it} \tag{4}$$

Energy poverty refers to people's inability to access sufficient energy due to economic, social, or geographical factors, which affects their quality of life and health. This may include a lack of electricity, fuel, and other basic energy sources. By using energy poverty as a moderating variable, the effects of different environmental regulations on social

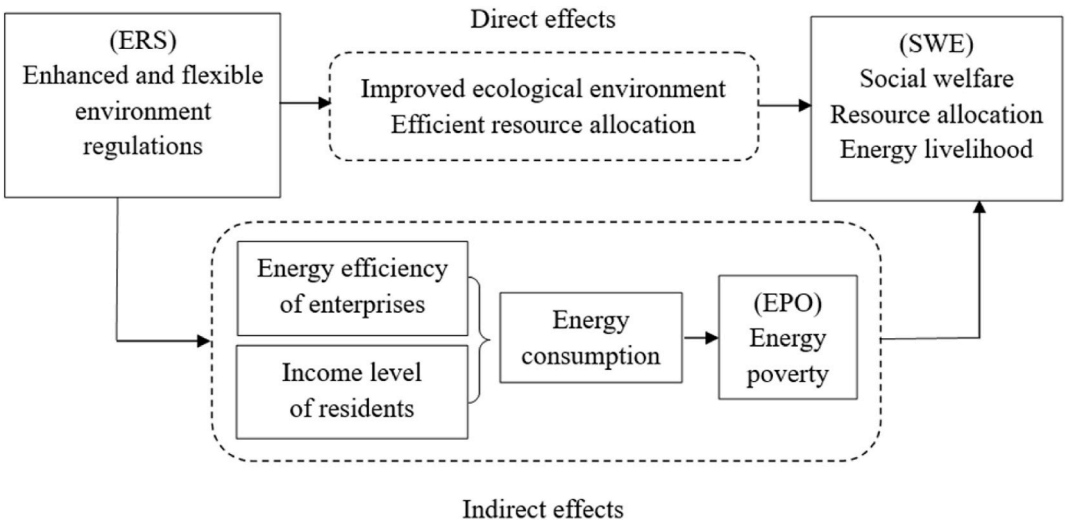

Fig. 2. The influence mechanism of ERS and EPO on SWE.

welfare would be more comprehensively evaluated. Environmental regulations may affect income levels and energy demand through mandatory regulations or residents' conscious environmental awareness, thereby having heterogeneous effects on social welfare. Some studies have assessed the impact of energy poverty and how it affects the social well-being of the people of Pakistan, including family, education, and economic development [69]. A household's energy needs are closely related to the type of fuel used and their limited of excessively priced renewable energy. Members of energy poverty households are more likely to suffer from respiratory diseases, fewer employment opportunities, and lower social benefits.

4.2. Variable measurement

4.2.1. Social welfare

In the Human Development Report, the United Nations Development Programme (UNDP) proposed Human Development Index (HDI), which measures the level of social welfare with three basic variables: Life, Health and Education [70]. Therefore, this paper constructs an indicator system of social welfare level from three dimensions: quality of life, state of health and level of education, as shown in Table 1.

Principal component analysis uses the idea of reducing dimensions to transform multiple indicators into fewer comprehensive indicators on the premise of losing less information. Therefore, the principal components analysis (PCA) method is adopted in this paper to measure the level of social welfare more reasonably and scientifically [46]. The basic form of this model is model (5):

$$\begin{Bmatrix} F_1 = u_{11}x_1 + u_{12}x_2 + \cdots + u_{1j}x_j \\ F_2 = u_{21}x_1 + u_{22}x_2 + \cdots + u_{2j}x_j \\ \vdots \\ F_m = u_{p1}x_1 + u_{p2}x_2 + \cdots + u_{pj}x_j \end{Bmatrix} \tag{5}$$

KMO (Kaiser Meyer Olkin) test statistic is an indicator used to compare the simple correlation coefficient and partial correlation coefficient between variables before factor analysis. KMO test and Bartlett sphericity test are conducted, as shown in Table 2. The KMO value is 0.797, which is greater than 0.5. The p-value of Bartlett is significant, indicating that it is suitable for principal component analysis. After forming the factor load matrix, the values of uniqueness are less than 0.6, so there is no need to eliminate variables. Then the scores of the principal components (f1, f2 and f3) are calculated by using the software of Stata 16. The principal component is a linear combination of indicator variables, that is, the coefficient of the linear combination of indicator variable data, and the principal component score would be calculated.

Table 1
Indicator system of social welfare level.

Primary indicators	Secondary indicators	Descriptions
Quality of life	GDP per capita (+)	Gross domestic product per capita
	GDP growth index per capita (+)	Gross domestic product growth index
	Per capita consumption expenditure of urban residents (+)	Consumption expenditure of urban residents
	Income gap between urban and rural areas (−)	Income of urban residents divided by income of rural residents
State of health	Number of medical and health institutions (+)	Number of Medical and health institutions
	Proportion of medical and health expenditure (+)	Medical and health expenditure divided by total financial expenditure
	Number of beds in medical institutions per 1000 people (+)	Number of beds in medical institutions
	Mortality rate of population (−)	Mortality rate of population
Level of education	Proportion of education expenditure (+)	Education expenditure divided by total financial expenditure
	Public library collection (+)	Public library collection
	Average years of education (+)	Years of education
	Proportion of people in higher education (+)	people in higher education

Table 2
The KMO test and Bartlett sphericity test.

Method	Indicator	Result
KMO test	KMO	0.797
Bartlett test of sphericity	χ^2	4627.926
	Degrees of freedom	66
	P-value	0.000

Finally, comprehensive indicators are generated based on the proportion of the variance contribution rate of principal components to the cumulative variance contribution rate. Among them, variance contribution rate refers to the proportion of variance explained by a principal component to the total variance. The larger this value, the stronger the ability of the principal component to integrate information from the original variables. The results show that the cumulative variance contribution rate is 0.7152, and the variance contribution rates of the

principal components are 0.4483, 0.1741 and 0.0928, respectively. Then the comprehensive indicators of social welfare are measured, as shown in equation (6).

$$SWE = \left(\frac{0.4483}{0.7152}\right) \times f_1 + \left(\frac{0.1741}{0.7152}\right) \times f_2 + \left(\frac{0.0928}{0.7152}\right) \times f_3 \tag{6}$$

4.2.2. Environmental regulations

By sorting out previous literature, the environmental regulations measured in this paper includes enhanced environmental regulation and flexible environmental regulation.

(1) Enhanced environmental regulation

Enhanced environmental regulation refers to the formulation of relevant laws by government departments, carrying out environmental protection supervision, and formulating enterprise pollutant discharge standards to protect the environment and reduce resource risks. At present, there is no clear and unified indicator for measuring enhanced environmental regulation (EER). Existing researches for measurement mainly include the number of environmental regulations [71], the proportion of pollution treatment expenditure to output [72], the pollution fees [73] and the energy efficiency ratios under different conditions [74]. Based on existing data and considering unbalanced development, this study measures the level of investment in environmental pollution control, which is the total investment in industrial pollution governance divided by GDP [75]. Enhanced environmental regulation is implemented by government departments with the aim of environmental protection and green ecology, and are social regulations for environmental governance through standardized pollution control standards. On the one hand, the traditional neoclassical theory holds that the opportunity cost of environmental governance is high, the production cost of regulated enterprises is increased, social productivity is reduced, and it has a direct impact on social welfare. On the other hand, the reasonable environmental investment also effectively motivates enterprises to improve their investment structure, adopt scientific and efficient green production methods, and promote the green development of ecological society. Therefore, this paper focuses on measuring enhanced environmental regulations through investment in environmental pollution control, and further studies the impact on social welfare. According to the measurement standards of ''China Statistical Yearbook on Environment'', it includes the sum of urban environmental infrastructure investment, industrial pollution source control investment, and "Three Simultaneous" project environmental protection investment, which are funds used to form fixed assets. The industrial pollution source control includes investment in the treatment of industrial exhaust gas, industrial wastewater, solid waste, etc. Environmental protection facilities in the "Three Simultaneous" construction project would be designed, constructed, and put into operation simultaneously with the main project, which is a green environmental protection policy.

(2) Flexible environmental regulation

Pargal & Wheeler [76] first proposed the concept of Flexible Environmental Regulation (FER). It refers to environmental organizations or other forms spontaneously formed by the public, which improve the environment and quality of life through negotiations with polluting enterprises, pressure on local governments and polluting enterprises, and other means. When enhanced environmental regulations implemented by the government are ineffective, environmental organizations or other spontaneous forces composed of the public take environmental protection actions spontaneously to improve the living environment and quality of life. According to previous research, on the one hand, the environmental awareness of residents increases with the improvement of population quality, that is, the education level regulates people's conscious behavior of protecting the ecological environment. On the other hand, with the improvement of living standards, the income level of residents to a certain extent affects the changes in lifestyle and the use of energy, including digital intelligence, green and clean. Therefore, this paper focuses on income level and education level as the main indicator dimensions. Then, the composite index is constructed, and the intensity of FER is measured using the entropy weighting method, with higher values indicating stricter FER [77,78]. In addition, the level of education received is measured by the average length of education, and the income level is measured by the average salary of urban workers. The information entropy model is applied to calculate the comprehensive indicator values in this paper [79]. Flexible environmental regulation represents the public's awareness of environmental protection, and its intensity increases with the improvement of residents' income and education level. The stronger the intensity, the better the public would perform in protecting the environment and saving resources.

4.2.3. Energy poverty

There are many international methods to measure the energy poverty line, mainly including physical quantity method, proportion method, expenditure method and basic demand method. The first three methods are subjective and arbitrary, so this paper adopts the basic demand method, which not only considers the impact of expenditure on energy demand, but also considers the impact of other exogenous factors. Based on energy demand, energy poverty is defined as a household energy shortage that cannot meet basic living needs, and there is a minimum energy demand representing the energy poverty line. By establishing an extended linear expenditure system model (ELES), the poverty line of energy consumption expenditure is calculated to reflect the energy poverty of each province [80]. The advantages of this model are simple operation and with low data requirements. In addition, due to the great differences in economic development, factor resources and other aspects in different regions of China, this paper divides the country into the eastern, central and western regions, and calculates the energy poverty lines of the three regions respectively. The energy poverty line is calculated as shown in equation (7):

$$P_i D_i = P_i d_i + \beta_i \left(I - \sum_{j=1}^{n} P_j d_j \right) \tag{7}$$

Where, P_i and P_j represent the prices of the i th and j th commodities respectively. D_i represents the actual demand of the i th commodity. d_i and d_j represent the basic demand of the i th and j th commodities respectively. I represents the per capita disposable income of residents. β_i represents marginal propensity to consume, $1 - \sum \beta_i$ indicates marginal propensity to save.

Because $P_i q_i$ and $\sum_{j=1}^{n} P_j q_j$ are constant terms, so that $\alpha_i = P_i q_i - \beta_i \sum_{j=1}^{n} P_j q_j$, $C_i = P_i Q_i$. Equation (7) is transformed as follows:

$$C_i = \alpha_i + \beta_i I + \delta_{it} \tag{8}$$

Where, C_i represents the energy consumption expenditure of residents, and I denotes the per capita disposable income of residents. α_i and β_i are the estimated coefficients of least squares regression (OLS). Equation (8) represents the energy poverty line, that is, the basic demand of per capita household energy consumption expenditure. In other words, it is the basic energy demand of households.

Due to the breadth of energy poverty, which refers to the incidence rate of energy poverty, it mainly indicates the possibility of residents falling into energy poverty and cannot reflect the intensity of energy poverty well. However, the depth of energy poverty, also known as the intensity of energy poverty, measures the size of the gap between energy poor households and the energy poverty line. The greater the intensity of energy poverty, the more severe the degree of energy poverty. Therefore, energy poverty intensity is measured by equation (9). $Consumption_i$ stands for per capital energy consumption expenditure, $Poverty_i$ stands for energy poverty line, that is, the basic energy demand of the

household is calculated from equation (8). When a province belongs to energy poverty, the energy poverty intensity is the absolute value of the energy consumption expenditure gap divided by the energy poverty line. When a province does not belong to energy poverty, the intensity of energy poverty is 0.

$$EP_i = \left| \frac{Consumption_i - Poverty_i}{Poverty_i} \right| \tag{9}$$

4.2.4. Control variable

By referring to existing relevant literature [81,82], this paper selects the level of industrial structure (IS), technological innovation (TI), government financial expenditure (FP), and education level (ED) as control variables, in order to control the impact of other variables on social welfare. The variables and definitions involved in the empirical analysis are shown in Table 3.

The relationship between different renewable energy sources, the role of technology, environmental protection and economic growth has been studied by using panel annual datasets of selected BRICS countries from 1997 to 2019 [83]. As household income inequality decreases, the use of renewable energy would increase. At the same time, Xue et al. [84] explored the asymmetric effects of renewable energy consumption, economic growth, and financial development on carbon emissions in emerging economies. The consumption of information, communication technologies and renewable energy sources hinders carbon emissions, Economic growth and population also contribute to the reduction of carbon emissions.

4.3. Descriptive statistics

Table 4 shows the results of descriptive statistical analysis for each variable. The minimum comprehensive level of social welfare is −1.970, and the maximum is 1.896. The average value is 0.006, which shows that the comprehensive level of social welfare in provinces is obviously different. The standard deviation for flexible environmental regulation is 0.185, slightly larger than the standard deviation of 0.119 for enhanced environmental regulation. In addition, the minimum and maximum values of energy poverty are 0 and 1, indicating that there are provinces with both energy poverty and non energy poverty. The standard deviations of technological innovation and education are 1.871 and 1.250, respectively, indicating that there is a relatively large gap in the level of technological innovation and education between different provinces, while the gap in industrial structure and financial expenditure is relatively small.

Table 3
The definition of variable.

Variables	Abbreviation	Measure
Explained Variable		
Social Welfare	SWE	Comprehensive level value calculated by PCA model (Three dimensions of life, health and education)
Explanatory Variable		
Enhanced environmental regulation	EER	Total investment in industrial pollution governance divided by GDP
Flexible environmental regulation	FER	Comprehensive index value calculated by entropy model (average years of education, average wages of urban workers)
Energy Poverty	EPO	Energy poverty line and energy poverty intensity are calculated by ELES model
Control Variable		
Industrial structure	IS	Output value of secondary industry divided by GDP
Technological innovation	TI	Total number of patents granted
Financial expenditure	FP	Government expenditure divided by GDP
Education	ED	Average years of education

Table 4
Summary of the Sample selection.

Variables	Obs	Mean	Std.Dev.	Min	Max
SWE	540	0.006	0.692	−1.970	1.896
EER	540	0.115	0.119	0.0000321	0.780
FER	540	0.308	0.185	0.0354	0.984
EPO	540	0.233	0.423	0	1
IS	540	0.424	0.0850	0.158	0.620
TI	540	9.177	1.871	2.773	13.47
FP	540	0.254	0.191	0.0842	1.354
ED	540	8.636	1.250	3.738	12.78

It shows the intensity of EER and FER in different provinces in 2020 (see Fig. 3). For enhanced environmental regulation, the intensity is higher in the eastern and northern regions, while the intensity is lower in the northeastern and southwestern regions. However, the intensity of flexible environmental regulation in the northwest and south regions is relatively high, while the intensity in the central region is relatively low.

5. Results

5.1. Data sources

Firstly, considering the consistency and comparability of indicator data, the time interval selected in this paper is 2003–2020. Due to the availability of data, Tibet, Hong Kong, Macao and Taiwan are excluded. Therefore, this paper selects the statistical data of 30 provinces in China, with a total of 540 samples. Secondly, the data of SWE and control variables are from the *China Statistical Yearbook* and provincial Statistical Yearbooks. The data of ERS and EER come from the *China Statistical Yearbook on Environment*, and the data of FER comes from the *China Statistical Yearbook*. For EPO, the data of per capita disposable income (I) comes from the *China Statistical Yearbook*, the data of energy price (P) comes from *WIND*, and the data of basic energy demand (Q) comes from the provincial Statistical Yearbooks. In addition, the empirical software used in this paper are Stata 16 and ArcMap 10.8.1.

5.2. Analysis of energy poverty

Considering the availability of micro data on household energy consumption and energy prices, oil expenditure, natural gas expenditure, electricity expenditure and other energy expenditure are selected to represent the energy consumption expenditure of households [5]. In addition, due to the great difference in the development level of economy, life and resources in China's regions, the energy utilization of residents varies significantly in different regions, and the use of the only energy poverty line cannot accurately determine the energy poverty situation. Therefore, this paper calculates the energy poverty lines of eastern, central and western regions in China respectively.

According to equation (7), the basic demand of residents' energy consumption expenditure in China and the three regions is calculated. Table 5 shows the estimated results. First of all, the national energy consumption poverty is 444.546 yuan per year. Secondly, for the three major regions, the energy consumption poverty in the eastern region is 503.381, which is higher than the national energy consumption level. Then the energy consumption poverty in the central region is 442.758, which is the closest one to the national energy consumption level. The energy consumption poverty in the western region is 359.461, which is lower than the national energy consumption level. More importantly, there are subtle differences in oil expenditure and natural gas expenditure among the eastern, central and western regions. However, the electricity expenditure in the three regions is obviously different. The electricity expenditure coefficients of the eastern and central regions are 169.721 and 120.401 respectively. The electricity expenditure coefficient of the western region is 82.628, which is significantly lower than

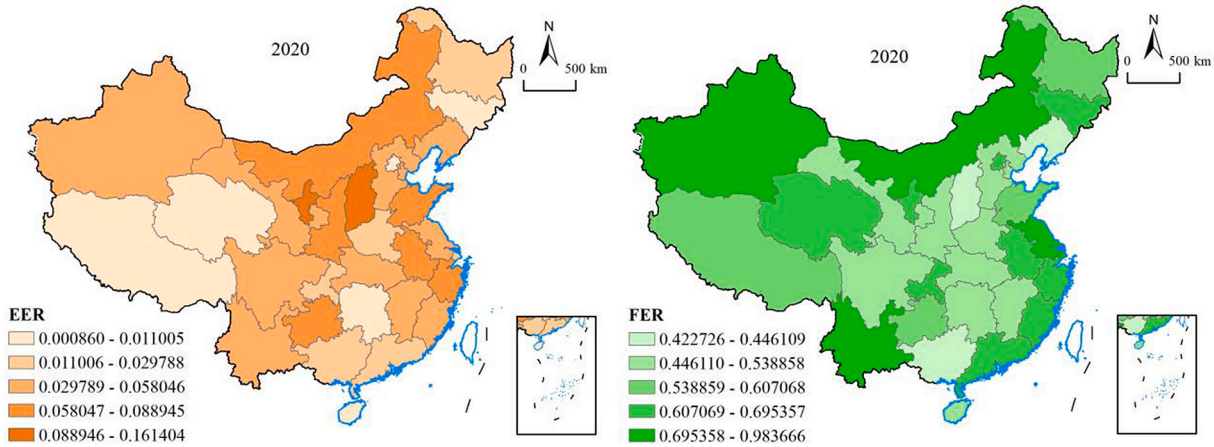

Fig. 3. The intensity of environmental regulations in 2020.

Table 5
The situation of energy poverty.

Indicators	All		East		Middle		West	
Parameter	α_i	β_i	α_i	β_i	α_i	β_i	α_i	β_i
Oil expenditure	67.823[b]	0.120[a]	97.953[a]	0.126[a]	78.310[a]	0.097[a]	47.658[a]	0.130[a]
	(12.157)	(0.007)	(35.503)	(0.013)	(17.727)	(0.011)	(12.716)	(0.08)
Natural gas expenditure	51.314[a]	0.038[a]	68.933[a]	0.034[a]	41.413[a]	0.027[a]	39.405[b]	0.057[a]
	(6.619)	(0.003)	(11.377)	(0.004)	(4.863)	(0.003)	(15.356)	(0.01)
Electricity expenditure	146.045[a]	0.115[a]	169.721[a]	0.101[a]	120.401[a]	0.116[a]	82.628[b]	0.132[a]
	(14.763)	(0.007)	(10.844)	(0.004)	(13.636)	(0.008)	(37.791)	(0.025)
Other energy expenditure	179.364[a]	−0.011[b]	166.774[a]	−0.080[b]	202.634[a]	−0.007[b]	189.770[a]	0.060[a]
	(10.677)	(0.051)	(11.098)	(0.004)	(20.725)	(0.003)	(23.077)	(0.015)
Total	444.546	0.262	503.381	0.253	442.758	0.233	359.461	0.379

Note: The standard deviation is shown in brackets.
[a] $p < 0.01$.
[b] $p < 0.05$.

that of the eastern and central regions. However, the other energy expenditure coefficient of the central region is 202.634, higher than that of the eastern and western regions. It can be seen that residents' energy consumption expenditure is obviously different in different regions, and the energy poverty level in the central region is closer to that of the whole country (see Fig. 4).

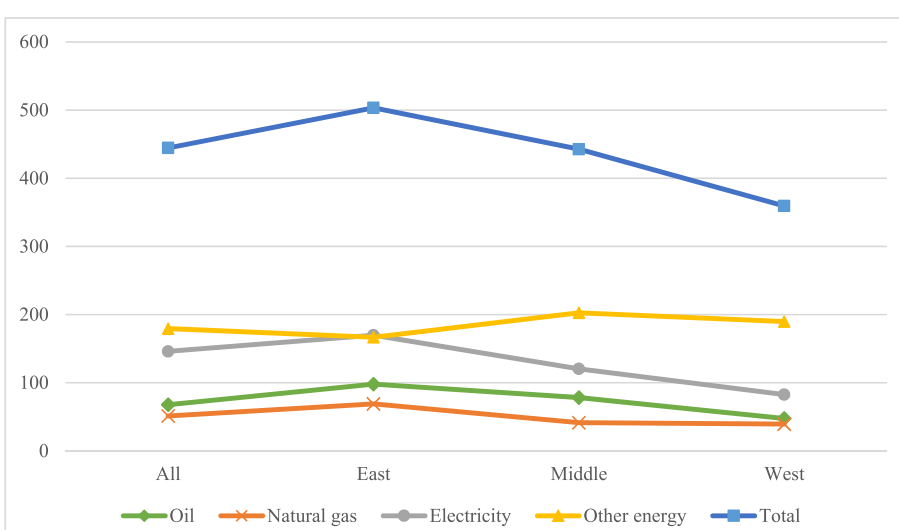

Fig. 4. The energy expenditure of different provinces in China.

5.3. Analysis of ERS and SWE

Table 6 shows the regression results of SWE affected by two types of environmental regulations. Groups 1 and 2 without control variables, and Groups 3 and 4 with control variables. Firstly, in Group 1, for FE and SYS-GMM, the regression coefficients of EER are −3.273 and −2.996 respectively, which are significantly negative at the 1 % level. However, the regression coefficients of FER are 2.834 and 3.293 respectively, which are significantly positive at the level of 1 %. This shows that EER has a negative correlation with SWE without control variables, but FER has a positive correlation with SWE. Secondly, control variables are added to Group 2. For FE, the regression coefficient of EER is −0.338 at the 1 % significant level. However, for FE, the regression coefficient of FER is 0.965, which is positive at the 1 % significant level. This shows that EER still has a negative impact on SWE, while FER has a significant positive impact on SWE with control variables (see Fig. 5).

On the one hand, this may be due to the "regression effect hypothesis" of enhanced environmental regulation, which believes that enhanced environmental regulation may increase the additional cost of emission reduction and pollution control of enterprises, thus reducing the competitiveness of enterprises [85], which may have a certain impact on the improvement of social welfare. On the other hand, flexible environmental regulation with the public, media and social groups as the main force may require enterprises to make green transformation and development. It would promote energy conservation and emission reduction of polluting enterprises, and gradually enhance the awareness of environmental protection, which would significantly boost the development of social welfare. However, for FE and SYS-GMM, the regression coefficients of IS are both negative, that are −0.669, −0.795, −0.212, −3.296, respectively. It indicates that the adjustment of industrial structure may inhibit the improvement of social welfare to some extent. For FE and SYS-GMM, the regression coefficients of TI are positive at the 1 % significant level, which are 0.356, 0.440, 0.268 and 0.433, respectively. For SYS-GMM, the regression coefficients of FP are 0.987 and 0.945 respectively, which are positive at the 1 % significant level, and the absolute value of the regression coefficients are relatively large. For FE, the regression coefficients of ED are 0.229 and 0.146 respectively, which are positive at the 1 % significant level. This shows that the improvement of technological innovation level, government financial expenditure level and education level could significantly promote the improvement of social welfare. In addition, the Hausman results of Group 1, Group 2, Group 3 and Group 4 are 20.80, 33.33, 58.07 and 20.95 respectively, which are at the 1 % significant level.

5.4. Analysis of EPO and SWE

Table 7 shows the regression results of EPO to SWE. In Group 1 excluding control variables, for FE and SYS-GMM, the regression coefficients of EPO to SWE are −0.984 and −1.015 respectively, which are negative at the 1 % significant level. Moreover, in Group 2 with control variables, for FE and SYS-GMM, the regression coefficients of EPO to SWE are −0.183 and −0.079 respectively, which are also negative at the 1 % significant level. It can be seen that EPO and SWE have negative correlation, and the occurrence of energy poverty may have a certain negative impact on the improvement of social welfare. This is because the incidence of energy poverty may make residents unable to obtain sufficient clean energy and increase the use of traditional energy. On the one hand, this may cause pollution to the ecological environment, damage the health of residents, reduce work efficiency and wage income. On the other hand, it may reduce leisure time and consumption of residents, thus reducing their happiness and damaging social welfare. For FE and SYS-GMM, the regression coefficients of TI on SWE are 0.314 and 0.453 respectively, which are positive at the 1 % significant level. The regression coefficient of ED is 0.228 at the 1 % level for FE. The level of technological innovation and educational development have significantly improved the level of social welfare. In addition, the Hausman results of Group 1 and Group 2 are 4.71 and 55.62 respectively.

5.5. Analysis of the moderating effect of EPO

Table 8 shows the regression results of the cross terms of ERS and EPO for SWE. Based on the control variables and EPO are introduced, Group 1 and Group 2 represent the influence of EER and FER on SWE respectively. Groups 3 and 4 represent the regression results of EER*EPO and FER*EPO on SWE respectively, that is, the moderating effect of EPO is in the impact of EER and FER on SWE. In Group 1 and Group 2, for FE and SYS-GMM, the regression coefficients of EER on SWE are −0.336 and −0.224 respectively. The regression coefficients of FER to SWE are 1.098 and 0.860 respectively, which are positive at the 1 % significant level. However, the regression coefficients of EPO on SWE are −0.182, −0.188, −0.212 and −0.538 respectively, which are negative at the 1 % significant level. This shows that EER has still negative correlation with SWE, FER has positive correlation with SWE, and EPO has significantly negative correlation with SWE, under the premise of introducing EPO and control variables. Then in Group 3, the regression coefficient of EER is −0.128, but not significant for FE. The regression coefficient of EER is −0.441 at the 1 % level for SYS-GMM. The regression coefficients of EPO are −0.058 and −0.078, respectively, but are not significant. For FE, the regression coefficient of EER*EPO on SWE is −0.812, which is

Table 6
The impact of ERS on SWE.

Variables	Group 1		Group 2		Group 3		Group4	
	FE	GMM	FE	GMM	FE	GMM	FE	GMM
EER	−3.273[a]	−2.996[a]			−0.338[a]	−0.156[c]		
FER			2.834[a]	3.293[a]			0.965[a]	−0.470[a]
IS					−0.669[a]	−0.795[b]	−0.212	−3.296[a]
TI					0.356[a]	0.440[a]	0.268[a]	0.433[a]
FP					−0.551[a]	0.987[a]	−0.544[a]	0.945[a]
ED					0.229[a]	−0.008	0.146[a]	−0.007
Cons	0.403[a]	0.372	−0.868[a]	−1.007[a]	−4.791[a]	−3.843[a]	−3.782[a]	−2.702[a]
R^2	0.249		0.856		0.895			
Hansen		0.760		0.757		0.994		0.994
Hausman	20.80[a]		33.33[a]		58.07[a]		20.95[a]	
AR(1)		0.042		0.080		0.006		0.094
AR(2)		0.186		0.132		0.913		0.535
Observation	540	540	540	540	540	540	540	540

Note.
[a] $p < 0.01$.
[b] $p < 0.05$.
[c] $p < 0.1$.

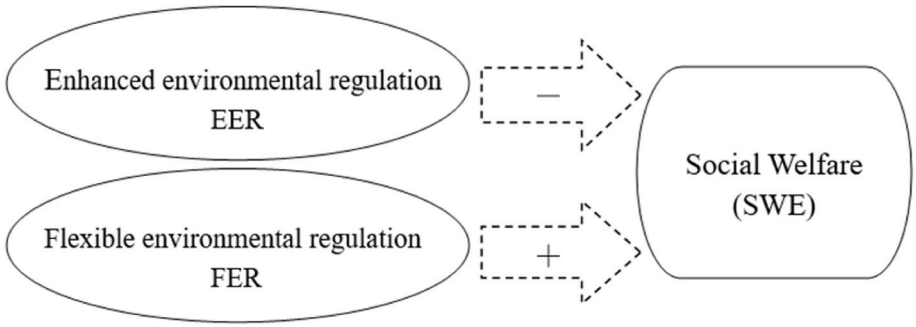

Fig. 5. The direct impact of EER and FER on SWE.

Table 7
The impact of EPO on SWE.

Variables	Group 1		Group 2	
	FE	GMM	FE	GMM
EPO	−0.984[a]	−1.015[a]	−0.183[a]	−0.079[a]
IS			−1.129[a]	−1.653[a]
TI			0.314[a]	0.453[a]
FP			−0.575[a]	0.414[a]
ED			0.228[a]	−0.069[a]
Cons	0.236[a]	0.243[a]	−4.181[a]	−2.942[a]
R^2	0.504		0.899	
Hansen		0.759		0.992
Hausman	4.71[b]		55.62[a]	
AR(1)		0.032		0.010
AR(2)		0.316		0.361
Observation	540	540	540	540

Note.
[a] $p < 0.01$.
[b] $p < 0.1$.

significant at the 1 % level. For SYS-GMM, the regression coefficient of EER*EPO on SWE is −0.656 at the 5 % significant level. Therefore, EER could inhibit SWE, and the absolute value of the coefficient of EER*EPO is greater than that of EER, and it is more significant. This shows that EPO has a positive moderating role in the influence of EER on SWE. Energy poverty aggravates the negative impact of enhanced environmental regulation on social welfare. This is because enhanced environmental regulation may bring about "compliance costs" and increase the cost of pollution control for enterprises. Too much consideration by enterprises on how to avoid pollution penalties imposed by government environmental regulations may limit the production scale of enterprises, which may reduce economic benefits and income levels. The incidence of energy poverty would intensify the negative impact of enhanced environmental regulation on social welfare. Energy poverty may lead to the low level of energy use of residents, squeezing labor time and occupying education time, which may slow down the pace of economic development and reduce the level of social welfare.

However, for FE in Group 4, the influence coefficient of FER on SWE is 1.111, which is positive at the 1 % significant level. For SYS-GMM, the regression coefficient of FER is 0.515, but it is not significant. The influence coefficients of EPO on SWE are −0.386 and −0.450 respectively. For FE, the influence coefficient of TER*EPO on SWE is 1.366, which is positive at the 1 % significant level. For SYS-GMM, the regression coefficient of FER*EPO is 1.546 at the 10 % significant level. Therefore, FER has a positive impact on SWE, while EPO has a negative impact on SWE. The absolute value of the coefficient of FER*EPO is greater than that of FER, and the regression results are more significant. This shows that EPO has a significant positive moderating role in the impact of FER on SWE. Energy poverty has significantly promoted the positive impact of flexible environmental regulation on the improvement of social welfare. This is because flexible environmental regulation promotes the green transformation of energy enterprises and the improvement of technological innovation level through reasonable public supervision and conscious environmental awareness. Even in energy poor areas, this would probably improve energy efficiency and make income and energy resource distribution more reasonable. In addition, residents

Table 8
The moderating effect of EPO.

Variables	Group 1		Group 2		Group 3		Group 4	
	FE	GMM	FE	GMM	FE	GMM	FE	GMM
EER	−0.336[a]	−0.224[b]			−0.128	−0.441[a]		
FER			1.098[a]	0.860[a]			1.111[a]	0.515
EPO	−0.182[a]	−0.188[a]	−0.212[a]	−0.538[a]	−0.058	−0.078	−0.386[a]	−0.450[b]
EER[c]EPO					−0.812[a]	−0.656[b]		
FER[c]EPO							1.366[a]	1.546[c]
IS	−1.047[a]	−1.509[a]	−0.574[a]	−2.207[a]	−1.082[a]	−1.623[a]	−0.801[a]	−1.493[c]
TI	0.302[a]	0.425[a]	0.198[a]	0.219[a]	0.308[a]	0.411[a]	0.188[a]	0.365[a]
FE	−0.530[a]	0.283[c]	−0.554[a]	−0.547[a]	−0.417[a]	0.199[c]	−0.658[a]	−0.061
ED	0.228[a]	−0.067[a]	0.125[a]	−0.049[a]	0.22[a]	−0.052[b]	0.122[a]	−0.078[c]
Cons	−4.074[a]	−2.670[a]	−2.794[a]	−0.716[a]	−4.103	−2.584[a]	−2.551[a]	−2.105[b]
R^2	0.902		0.911		0.905		0.913	
Hansen		0.993		0.543		0.980		0.988
Hausman	51.64[a]		17.98[b]		52.36[a]		18.53[b]	
AR(1)		0.007		0.018		0.004		0.007
AR(2)		0.298		0.890		0.264		0.270
Observation	540	540	540	540	540	540	540	540

Note.
[a] $p < 0.01$.
[b] $p < 0.05$.
[c] $p < 0.1$.

consciously strengthen their awareness of environmental protection and reduce the utilization of traditional energy, thus promoting the development of education and the improvement of social welfare.

5.6. Empirical testing

5.6.1. Two stage least square test

Based on relevant concepts, endogenous variables refer to explanatory variables that are related to the error term, while exogenous variables refer to explanatory variables that are not related to the error term. In addition to GMM estimation, the TSLS method could be used to solve endogeneity problems, a two-stage least squares regression. Endogeneity refers to the correlation between one or more explanatory variables in a model and random perturbation terms. The existence of endogenous problems would make the parameter estimation of the model imprecise, thus the endogenous problems must be taken into account when model selection is carried out.

The test results of two stage least square method is shown in Table 9. The L.EER and L.FER represent the second lag of enhanced environmental regulation and flexible environmental regulation, respectively, and the L.EPO represents the second lag of energy poverty. The regression results of the L.EER and L.FER on SWE with Group 1 and Group 2, and the regression results of the moderating effect with Group 3 and Group 4. The coefficients of L.EER and L.FER on SWE are −5.892 and 4.855 respectively, which are significant at the 1 % level. In the Group 3 and Group 4, the regression coefficients of L.EER and L.FER on SWE are −0.404 at the 10 % and 1.519 at the 1 % respectively. In addition, the moderating coefficients of L.EER*EPO and L.FER*EPO on SWE are −2.439 at the 1 % level and 10.780 at the 10 % level respectively. Therefore, the energy poverty has significant positive moderating effect. It can be seen that the test results of Two Stage Least Square are consistent with the above conclusions by using SYS-GMM. Therefore, the above regression results have passed the test of TSLS, and the conclusions are scientific and reliable.

5.6.2. Robustness results

This paper selects data from 2011 to 2020 as new sample data from all years for robustness test. Table 10 shows the regression results of the test. Group 1 and Group 2 without moderating variables, while Group 3 and Group 4 with moderating variables. Firstly, in Group 1, the regression coefficient of EER on SWE is −0.154, which is significant at the 1 % level. In Group 2, the regression coefficient of FER on SWE is 0.212 at the 1 % significant level. EER has a negative impact on SWE, while FER has a significant positive impact on SWE. The enhanced environmental regulation may increase the additional costs of emissions reduction and pollution control for enterprises, reduce the competitiveness, and thus affect the social welfare. However, the flexible environmental regulation requires enterprises to undergo green transformation and development, promotes energy conservation and emission reduction of polluting enterprises, and improves social welfare. It can be seen that the test results of Group 1 and Group 2 are consistent with the above conclusions. Secondly, in Group 3, the regression coefficient of EER is −0.223, which is significant at the 1 % level. The regression coefficient of EPO is −0.135, which is significant at the 10 % level. The influence coefficient of EER*EPO on SWE is −8.115, which is significant at the 1 % level. The absolute value of the coefficient of EER*EPO is still greater than that of the EER. The occurrence of energy poverty would exacerbate the negative impact of enhanced environmental regulation on social welfare, leading to low energy utilization levels among residents, squeezing labor and education time, and thus reducing social welfare levels. Thirdly, in Group 4, the regression coefficient of FER on SWE is 0.875, which is positive at the 1 % significant level. The regression coefficient of EPO is −1.721, which is significant at the 5 % level. In addition, the influence coefficient of FER*EPO on SWE is 7.189, which is positive at the 5 % significant level. The regression coefficient of FER*EPO is far greater than that of FER. It can be seen that the regression results of Group 3 and Group 4 are also consistent with the above conclusions. Therefore, the basic conclusion of this paper is robust.

Table 9
The test of Two Stage Least Square method.

Variables	Group 1	Group 2	Group 3	Group 4
	TSLS	TSLS	TSLS	TSLS
L.EER	−5.892[a]		−0.040[c]	
L.FER		4.855[a]		1.519[a]
L.EPO			−0.074	−1.221[b]
L.EER[c]EPO			−2.439[a]	
L.FER[c]EPO				10.780[c]
IS	−0.084	1.118[a]	−3.196[a]	−1.765[a]
TI	0.957[b]	−2.369[a]	0.214[a]	0.173[b]
FE	−0.014	0.714[a]	−0.616[b]	−1.160
ED	1.999[a]	−1.316[b]	0.119[b]	−0.019
Cons	−5.734[a]	8.600[a]	−1.642[a]	−0.855
F test	4.940[a]	12.680[a]	15.830[a]	3.300[a]
Observation	471	480	471	480

Note.
[a] $p < 0.01$.
[b] $p < 0.05$.
[c] $p < 0.1$.

Table 10
The results of robustness.

Variables	Group 1	Group 2	Group 3	Group 4
	GMM	GMM	GMM	GMM
EER	−0.154[a]		−0.223[a]	
FER		0.212[a]		0.875[a]
EPO			−0.135[c]	−1.721[b]
EER[c]EPO			−8.115[b]	
FER[c]EPO				7.189[b]
IS	−2.896[a]	−2.692[a]	−2.665[a]	−3.362[a]
TI	0.254[a]	0.295[a]	0.281[a]	0.073[a]
FE	−0.948[a]	−0.918[a]	−1.086[a]	−2.186[a]
ED	−0.126[a]	−0.148[a]	−0.148[a]	−0.161[a]
Cons	0.433[a]	0.089[c]	0.405	2.725[a]
Hansen	0.489	0.437	0.384	0.416
AR(1)	0.058	0.055	0.097	0.069
AR(2)	0.335	0.425	0.717	0.426
Observation	300	300	300	300

Note.
[a] $p < 0.01$.
[b] $p < 0.05$.
[c] $p < 0.1$.

5.7. Discussion

In order to intuitively express the main conclusions, after conducting the robustness test, a graphical summary is provided (see Fig. 6). Firstly, different types of ERS have different impacts on SWE. EER has a negative relationship with SWE, but FER has a positive relationship with SWE. This is the same as Wang and Zhang (2022), in terms of spatial effect, enhanced environmental regulation has a negative spatial spillover effect on the promotion of social welfare, while flexible environmental regulation has a positive spatial spillover effect. However, it is different from Xiao et al. [24]. They believe that enhanced environmental regulations may reduce the wages of employees in polluting enterprises, leading to decline in social welfare. Secondly, EPO has a negative relationship with SWE. This conclusion is consistent with Li et al. [46], which holds the same view that long-term energy poverty may reduce social welfare, and only access to energy efficiency will it benefit. Areas with low energy efficiency may aggravate energy poverty.

Moreover, EER*EPO has a negative relationship with SWE, but FER*EPO has a positive relationship with SWE. This shows that EPO has a significant positive moderating role in the influence of EER and FER on

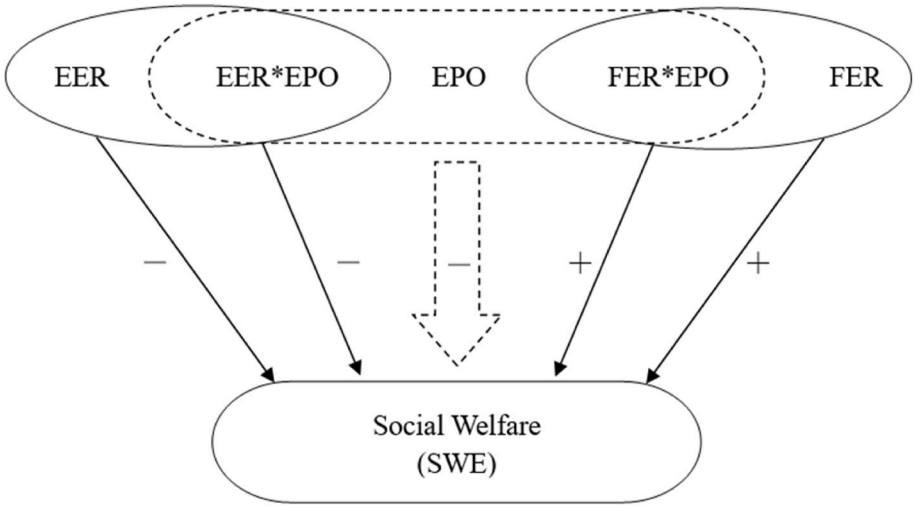

Fig. 6. The influence of ERS and EPO on SWE.

SWE. On the one hand, EPO aggravates the negative impact of enhanced environmental regulation on the improvement of social welfare. On the other hand, EPO promotes the positive impact of flexible environmental regulation on the improvement of social welfare. This is the same as the conclusion of Ma et al. [36], which believes that strict environmental regulations may lead to greater affordability problems for households using non-clean energy, deepen energy poverty, and inhibit the improvement of social welfare.

6. Conclusion and policy implications

6.1. Conclusion

This paper draws the conclusions as follows: First, enhanced environmental regulation has a significant negative impact on social welfare, while flexible environmental regulation has the opposite, which shows that flexible environmental regulation promotes the social welfare, especially in energy poverty regions. Flexible environmental regulation can stimulate the technical level of energy enterprises, and improve energy efficiency through public awareness of environmental protection, education and supervision, which will make the income of residents and the distribution of energy resources more reasonable. Second, energy poverty has a significant negative impact on social welfare, which indicates that the energy poverty inhibits the social welfare. Since the energy poverty makes residents unable to consume clean energy, which may pollute the environment and damage the health, thus reducing social welfare. Third, the energy poverty has significant positive moderating effect. That is, energy poverty not only aggravates the inhibitory effect of enhanced environmental regulation on social welfare, but also significantly promotes the impact of flexible environmental regulation on social welfare. Finally, the energy poverty varies in different regions. Specifically, the energy consumption capacity of the eastern region is relatively high, that of the western region is relatively low, and that of the central region is the same as that of the whole country. The eastern region has a higher level of economy, education and living than the central and western regions. While the energy infrastructure in the western region is weak, and the residents' income and education level are low, leading to different basic demands for energy consumption expenditure in different regions.

6.2. Policy implications

Based on the above conclusions, the following suggestions are put forward:

Enhanced environmental regulation has a negative impact on social welfare, while flexible environmental regulation has the opposite. On the one hand, the government should attach importance to the strength of environmental regulation, reasonably improve the construction of environmental governance policy system, establish corresponding supporting measures, and ensure the effective implementation of policies. The entry threshold for heavily polluting industries should be raised, and a number of green industrial manufacturing parks should be constructed. It would promote active cooperation among high polluting, high energy consuming, and high emission enterprises, thereby improving energy utilization efficiency and social welfare. On the other hand, the government should promote the concept of green, raise public awareness of environmental protection and awareness of energy poverty, guide people to gradually transition to clean and efficient renewable energy, and reduce the proportion of traditional energy consumption. It would force enterprises to exert subjective initiative and carry out green technology innovation. Based on the perspective of regional heterogeneity, the government maybe should maintain the regulatory intensity of heavy polluting enterprises in the eastern region, build industrial innovation platforms, and build an integrated mechanism of factor markets. In the central and western regions, the government should strengthen the attention of enterprises to pollution control, increase investment in research and development funds and personnel for pollution control. In addition, the government should improve the environmental regulation and governance system, promote green economic efficiency, alleviate energy poverty, and improve the level of social welfare.

The clean energy infrastructure in China is not complete, and the development and utilization rate of renewable energy is still low. Therefore, the government should optimize the energy structure, carry out energy transformation, strengthen the development and popularization of clean energy such as wind, solar, biomass, hydrogen and geothermal energy, and establish an energy system focusing on clean energy. Meanwhile, it should improve energy efficiency and reduce the proportion of traditional energy use. For example, the government can compensate the cost of fossil energy transformation through financial subsidies and tax relief, to promote clean energy infrastructure construction and renewable energy development. More importantly, environmental pollution and damage caused by renewable energy development and utilization should be strictly controlled to alleviate energy poverty and improve social welfare. In addition, due to unbalanced regional economic development and uneven education levels, there are significant differences in energy poverty in different regions of China. In view of energy poverty at different levels, regional

heterogeneity should be fully considered, and differentiated environmental regulation policies should be adopted to coordinate the energy, environment and economic development in the region, optimize the energy consumption structure, so as to alleviate energy poverty and improve social welfare.

CRediT authorship contribution statement

Jian Tang: Conceptualization, Methodology, Writing – review & editing. **Xiaoyue Huang:** Data curation, Writing – original draft, Software. **Hsing Hung Chen:** Software, Writing – original draft. **Wangbo Gao:** Visualization, Investigation. **Xia Liu:** Supervision, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Project of Social Sciences Research for Colleges and Universities in Henan Province in 2024(2025-ZZJH-148).

Data availability

Data will be made available on request.

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